

UNIT TWO Group Theory

LCM - Least common Multiple

GCD - Greatest Common Divisor.

Prime no :- A positive integer, $n \geq 2$ is said to be prime if n is divisible by 1 & itself only.

Co-Prime no :-

Let a & b be two integers said to be co-prime if $(a, b) = 1$

Dividend = Divisor $\times$ quotient + Remainder.

* Modulus arithmetic:-

$$a \equiv b \pmod{n}$$

Read $\Rightarrow$ a congruent to b mod n .

$$12 \equiv 02 \pmod{5}$$

$$\begin{array}{r} 2 \\ 5 \overline{) 12} \\ \underline{-10} \\ 2 \end{array}$$

IMP * Fermat Theorem :-

If p is prime number
 & $(a, p) = 1$ then $a^{p-1} \equiv 1 \pmod{p}$

* Group :-

Let G be any set with $*$ be the binary operation on G then $(G, *)$ is said to be group if it satisfy the following properties.

1. If $a, b \in G$ then $a * b \in G$ - (closure property)
2. If $a, b, c \in G$ &
 $a * (b * c) = (a * b) * c$ - (Associative)
3. If $e \in G$, such that $a * e = e * a = a$
 $e \in G$, e = identity element. (Identity prop)
4. If $a, b \in G$ such that $a * b = e = b * a$
 where e is identity element.
 b is called inverse element of a .

Ex. $(\mathbb{N}, \cdot)$ is group?

$\Rightarrow \mathbb{N} = \{1, 2, 3, 4, 5, 6, \dots\}$

$$1+2 \in \mathbb{N}$$

$$1 \cdot 2 = 2 \in \mathbb{N}$$

(i) $a, b \in \mathbb{N}$ then $a \cdot b \in \mathbb{N}$... It holds
closure property

$$a * (b * c) = (a * b) * c$$

$$a = 1 \quad b = 2 \quad c = 3$$

$$\text{L.H.S} = a * (b * c)$$

$$= 1 \cdot (2 \cdot 3)$$

$$= 1 \cdot 6$$

$$= 6$$

$$\text{R.H.S} = (a * b) * c$$

$$= (1 \cdot 2) \cdot 3$$

$$= 2 \cdot 3$$

$$= 6$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

(ii) $a, b, c \in \mathbb{N}$ then $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
(Associative hold)

$$e = 1$$

$$1 \cdot 1 = 1$$

$$2 \cdot 1 = 2$$

$$3 \cdot 1 = 3$$

Here $e = 1$, is identity element

(iii) It holds Identity property

(iv) if $a, b \in G$ such that

$$a * b = e = b * a$$

$$1 \cdot (1) = 1$$

$1 \cdot \left(\frac{1}{2}\right) = 1 \dots$ but $\frac{1}{2}$ does not exist in set
inverse element.

$\therefore \mathbb{Z}^+$ does not hold the property of
inverse element.

$(\mathbb{N}, \cdot)$ is Not group.

Ex. Find the p remainder when 7^{2001} divided by 5.

$\Rightarrow p = 5 \quad a = 7$

$$(a, p) = (7, 5) = 1$$

$\therefore$ by Fermat Theorem

$$a^{p-1} = 1 \pmod{p}$$

$$= 7^4 = 1 \pmod{5}$$

$$a \cdot (7)^{500} = (1)^{500} \pmod{5}$$

$$7^{2000} = 1 \pmod{5}$$

$$7 \cdot 7^{2000} = 7 \pmod{5}$$

$$7^{2001} = 7 \pmod{5}$$

$\therefore 7$ is greater than 5.
 $5 \nmid 7$

$$\therefore 7^{2001} = 2 \pmod{5}$$

$\therefore$ Remainder = 2

Ex: Find remainder when 3^{31} is div by 7
 or
 Find the value of $3^{31} \pmod{7}$

$$\Rightarrow p = 7 \quad a = 3$$

$$(a, p) = (3, 7) = 1$$

by fermat thm

$$a^{p-1} = 1 \pmod{p}$$

$$3^6 = 1 \pmod{7}$$

$$(3^6)^5 = 1^5 \pmod{7}$$

$$3^{30} = 1^5 \pmod{7}$$

$$3 \cdot 3^{30} = 3 \cdot 1 \pmod{7}$$

$$\text{remainder} = 3 \pmod{7}$$

$$\therefore \text{remainder} = 3$$

$$\begin{aligned}
 i^0 &= \sqrt{-1} \\
 i^2 &= -1 \\
 i^3 &= i^2 \cdot i = -i \\
 i^4 &= i^3 \cdot i = 1
 \end{aligned}$$

Imp.

Ex let $G = \{1, -1, i, -i\}$ be the set
then show that $(G, \cdot)$ is group.

$\Rightarrow$

$$G = \{1, -1, i, -i\}$$

* $\rightarrow \cdot \leftarrow$ Multiplication.

(1) closure

$\forall a, b \in G$ then $a * b \in G$

$$a, b \in G \text{ then } a \cdot b \in G$$

$$(1)(-1) = -1 \in G$$

$$(1)(i) = i \in G$$

$$(1)(-i) = -i \in G$$

$$(-1)(1) = -1 \in G$$

$$(-1)(-1) = 1 \in G$$

$$(i)(-i) = 1 \in G$$

G hold closure property.

2) Associativity :-

$\forall a, b, c \in G$ such that.

$$a * (b * c) = (a * b) * c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

let $a = -1, b = i, c = -i$

$$\text{L.H.S} = a(b \cdot c) \quad a = -1, b = i, c = -i$$

$$\begin{aligned} \text{L.H.S} &= a \cdot (b \cdot c) = (-1) [(i)(-i)] \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= (a \cdot b) \cdot c = [(-1)(i)] \cdot (-i) \\ &= -i \cdot -i \\ &= i^2 \\ &= -1 \end{aligned}$$

$$\text{R.H.S} = \text{L.H.S} \quad (-1) \cdot (-1)$$

G holds associative property.

3) Existence of Identity :-

$\forall a \in G, \exists e \in G$ such that

$$a * e = e * a = a$$

i.e. $a \cdot e = e \cdot a = a$

$$(1)(1) = 1$$

$$(-1)(1) = 1$$

$$(i)(1) = i$$

$$(-i)(1) = -i$$

$\therefore \boxed{e = 1}$ identity element of G .

4)

Existence of Inverse.

$\forall a, b \in G$ such that

$$a * b = b * a = e$$

$$\text{i.e. } a \cdot b = b \cdot a = 1$$

$$(1)(1) = 1$$

$$(-1)(-1) = 1$$

$$(i)(-i) = 1$$

$$(-i)(i) = 1$$

$\Rightarrow \forall a \in G \exists b \in G$ such that
 $a \cdot b = b \cdot a = e$

$\rightarrow (G, \cdot)$ is group.

Ex:-

* Abelian group / Commutative :-

$(G, *)$ is group is said to be abelian group if $\forall a, b \in G$ such that $a * b = b * a$

$$N = \{1, 2, 3, \dots\}$$

Set of natural no. under multiplication :-

$$\begin{matrix} 2 \times 3 = 6 \\ 3 \times 2 = 6 \end{matrix} \} \text{ same.}$$

It holds abelian property

Set of natural no. under subtraction :-

$$\begin{matrix} 2 - 3 = -1 \\ 3 - 2 = 1 \end{matrix}$$

It does not hold abelian prop.

* Show that $(G, \cdot)$ is abelian group :-

→ Here $G = \{1, -1, i, -i\}$

∴ Check if it is a group

Ans :- Note :- {last question closeness prop}

$$(1)(-1) = -1$$

$$(-1)(1) = -1$$

$$(i)(i) = -1$$

$$(i)(-i) = 1$$

$$(1)(i) = i$$

$$(i)(1) = i$$

$$(1)(-i) = -i$$

$$(-i)(1) = -i$$

$$(1)(i) = -i$$

$$(i)(-1) = -i$$

$$(i)(-i) = 1$$

$$(-i)(i) = 1$$

for $\forall a, b \in G$ then $a * b = b * a$
 G under multiplication is abelian grp.

Ex. In $\mathbb{Z}$, we define $a * b = a + b + 1$
 show that $(\mathbb{Z}, *)$ is an abelian grp.

$$\Rightarrow \mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

$$a * b = a + b + 1$$

$$\begin{aligned} (-3) * 1 &= -3 + 1 + 1 \\ &= -1 \end{aligned}$$

1] Closureness

$$\forall a, b \in G, a * b \in G$$

$$a, b \in \mathbb{Z} \rightarrow a * b \in \mathbb{Z}$$

$$a \cdot b = a + b + 1 \quad \forall a, b \in \mathbb{Z}$$

$\mathbb{Z}$ holds closure prop.

$$\begin{aligned} i &= (i) (1) \\ i- &= (1) (i-) \\ i+ &= (i) (1) \\ i- &= (i-) (i) \\ i &= (i-) (1) \\ i &= (i) (i-) \end{aligned}$$

$$a + 0 = a \quad \forall a \in \mathbb{Z}$$

$$a + b + 1 = b + a + 1$$

$$\begin{aligned} (a + b + 1) + 1 &= (b + a + 1) + 1 \\ a + b + 2 &= b + a + 2 \end{aligned}$$

$$a + b + 1 = b + a + 1$$

$$a + 0 = a \quad \forall a \in \mathbb{Z}$$

Ex:- $H = \{0, 2\}$ is subgroup of Z_4 under binary operation Addition module 4.

$\Rightarrow Z_4 = \{0, 1, 2, 3\}$
 $* \rightarrow + (\text{Add}^n)$

$H = \{0, 2\}$

Here $H \leq Z_4$.

1) Closeness prop:-
 $0 + 2 = 2 \in H$

2) Associativity
 This is obviously associative.

3) Existence of Identity.
 $0 + 0 = 0$
 $2 + 0 = 2$

$\Rightarrow e = 0$ Identity element exist.

4) Existence of Inverse
 $0 + 0 = 0$
 $2 + 2 = 4 = 0$

Inverse exist for all element of H .

$H = \{0, 2\}$ is subgroup of Z_4

* In $(\mathbb{Z}_4, +)$,

$\{0\} + \{0, 1, 2, 3\}$ --- both are trivial subgroup.

$\{0, 2\} \rightarrow$ Example of Non-trivial subgroup of $\mathbb{Z}_4$.

12. Let $\{x_n\}$ be a sequence of real numbers such that $x_n \in (-1, 1)$ for all n . Define $y_n = \frac{x_n + 1}{2}$. Show that $\{y_n\}$ is a sequence of real numbers such that $y_n \in (0, 1)$ for all n .

$\{x_n\} = \{1, 0, -1, 0, 1, -1, 0, 1, -1, 0, \dots\}$

Then $y_n = \frac{x_n + 1}{2}$

$$\begin{aligned} 1 &= 1 \Rightarrow y_1 = \frac{1+1}{2} = 1 \\ 0 &= 0 \Rightarrow y_2 = \frac{0+1}{2} = \frac{1}{2} \\ -1 &= -1 \Rightarrow y_3 = \frac{-1+1}{2} = 0 \\ 0 &= 0 \Rightarrow y_4 = \frac{0+1}{2} = \frac{1}{2} \\ 1 &= 1 \Rightarrow y_5 = \frac{1+1}{2} = 1 \\ -1 &= -1 \Rightarrow y_6 = \frac{-1+1}{2} = 0 \end{aligned}$$

$\{y_n\}$ is a sequence of real numbers such that $y_n \in (0, 1)$ for all n .

$$\begin{aligned} 1 &= 1 \Rightarrow y_1 = \frac{1+1}{2} = 1 \\ 0 &= 0 \Rightarrow y_2 = \frac{0+1}{2} = \frac{1}{2} \\ -1 &= -1 \Rightarrow y_3 = \frac{-1+1}{2} = 0 \\ 0 &= 0 \Rightarrow y_4 = \frac{0+1}{2} = \frac{1}{2} \\ 1 &= 1 \Rightarrow y_5 = \frac{1+1}{2} = 1 \\ -1 &= -1 \Rightarrow y_6 = \frac{-1+1}{2} = 0 \end{aligned}$$

$$a^n = n.a$$

Ex. $(\mathbb{Z}, +)$ is cyclic? if yes, find its generator?

$$\Rightarrow \mathbb{Z} = \{ \dots -3, -2, -1, 0, 1, 2, 3, \dots \}$$

$\leftarrow$ Binary operation.

$$1^1 = n.a = 1.1 = 1$$

$$1^2 = 2.1 = 2$$

$$1^3 = 3.1 = 3$$

$$1^0 = 0.1 = 0$$

$$1^{-1} = (-1).1 = -1$$

$$1^{-2} = (-2).1 = -2.$$

1 is generator of $\mathbb{Z}$.

$$(-1)^1 = -1$$

$$(-1)^2 = 2(-1) = -2$$

$$(-1)^3 = 3(-1) = -3$$

$\vdots$

$$(-1)^0 = 0(-1) = 0$$

$$(-1)^{-1} = (-1)(-1) = 1$$

$$(-1)^{-2} = (-2)(-1) = 2$$

Page No.
 Date

$\Rightarrow -1$ is also generator of $\mathbb{Z}$.

* Subgroup:-

Let $(G, *)$ is a group and $H \subseteq G$
It is called subgroup of G , if H itself
is a group.

Ex:- $(G, \cdot)$ Here $G = \{1, -1, i, -i\}$

$$\& H = \{1, i, -i\}$$

(1) Closurenness

$$\forall a, b,$$

$$1(i) = i \in H$$

$$1(-i) = -i \in H$$

$$i(-i) = 1 \in H$$

$\therefore$ It holds closurenness property.

(2) Associative

$$a = 1 \quad b = i \quad c = -i,$$

$$\begin{aligned} \text{L.H.S} &= a.(b.c) \\ &= 1.(i.-i) \\ &= 1.1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= (a.b).c = (1.i).-i \\ &= i.-i \\ &= 1 \end{aligned}$$

$\mathbb{Z}_4$ holds associative property.

3.) Existence of Identity :-
 $a * e = e * a = a$

$$1(1) = 1$$

$$-1(1) = -1$$

$$1(1) = 1$$

$$1(1) = 1$$

$$-1(1) = -1$$

$$1(-1) = 1$$

$e = 1 \rightarrow$ Identity ele of $\mathbb{Z}_4$

4.) Existence of Inverse.

$$1(1) = 1$$

$$-1(i) = 1$$

$$1(-i) = 1$$

$\therefore \mathbb{Z}_4$ holds existence of inverse prop.

* CO-set :-

Let $(G, *)$ be a group of $(H, *)$
 be the subgroup of G .

for any $a \in G$, the set

Page No.
 Date

$aH = \{ a * h \mid h \in H \}$ is called left co-set of H & determined by a .

Similarly,

$Ha = \{ h * a \mid h \in H \}$ is called right co-set of H & determined by a .

Ex:- Let $G = (\mathbb{Z}_4, +)$ & $H = \{0, 2\}$ is subgroup of $(\mathbb{Z}_4, +)$ then find its left side & right co-set of H in G .

$\Rightarrow (G, *)$ — group of $(H, *)$ is subgroup of G & $a \in G$ then, left co-set of H ,

$$aH = \{ a * h \mid h \in H \}$$

Here, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ & $H = \{0, 2\}$

$$aH = \{ a + h \mid h \in H \}$$

$$a = 0,$$

$$\begin{aligned} 0H &= \{ 0 + 0 = 0, 0 + 2 = 2 \} \\ &= \{ 0, 2 \} = H \end{aligned}$$

$$aH = \{ 0, 2 \}$$

$$1H = \{1, 3\}$$

$$2H = \{2, 0\}$$

$$3H = \{3, 1\}$$

$$0H = 2H = \{0, 2\}$$

$$1H = 3H = \{1, 3\}$$

We got here, two distinct coset.

Now, right coset is,

$$Ha = \{h + a \mid h \in H\} \quad (*, 0)$$

$$a = 0.$$

$$H_0 = \{0 + 0 = 0, 2 + 0 = 2\}$$

$$H_0 = \{0, 2\}$$

$$H_1 = \{1, 3\}$$

$$H_2 = \{2, 0\}$$

$$H_3 = \{3, 1\}$$

$$\text{Here } H_0 = H_2 = \{0, 2\}$$

$$H_1 = H_3 = \{1, 3\}$$

$\Rightarrow$ we got two distinct right coset